

PRABU JAYANT

+91 8904261616 ◇ Bangalore, Karnataka, India

prabu.jayant2022@gmail.com ◇ linkedin.com/in/prabu-jayant ◇ github.com/prabujayant ◇ prabujayant.vercel.app

RESEARCH INTERESTS

Machine learning and deep learning; transformer-based architectures and representation learning; anomaly detection and classification in high-dimensional data; applied ML for cybersecurity, network analytics, and intelligent systems.

EDUCATION

RV College of Engineering, Bangalore 2022 – 2026
B.E. Computer Science and Engineering (Cybersecurity Specialization) CGPA: 8.87 / 10.0
Relevant Coursework: Operating Systems, Computer Networks, Data Structures & Algorithms, System Design, Database Systems, Machine Learning, Applied Statistics

PUBLICATIONS

Co-author, 5 peer-reviewed publications (2024–2025) spanning *IEEE Access* and IEEE international conferences, on applied machine learning for encrypted SaaS traffic classification, network intrusion detection, and IoT anomaly detection. Total citations: 34; h-index: 2 (Google Scholar).

1. A. Ravi, B. Jnyanadeep, M. V. Gagana, P. Jayant, A. Pranav, and P. Siddappa, “CASB Security Analytics for Encrypted SaaS Traffic: A Hybrid Transformer-Based Classification Framework in Enterprise Cloud Ecosystems,” *IEEE Access*, vol. 13, pp. 216318–216339, 2025.
2. P. Jayant, E. Vincent, M. Moharir, and A. K. A. R., “Smart Health Monitoring and Anomaly Detection Using Internet of Things (IoT) and Artificial Intelligence,” in *Proc. 2nd Int. Conf. Intelligent Cyber Physical Systems (ICICPS)*, 2024. (26 citations)
3. P. Jayant, M. P. Shetty, S. Jeevan, M. Moharir, and A. R. A. Kumar, “Intrusion Detection in Network Traffic Using LSTM and Deep Learning,” in *Proc. 15th Int. Conf. Computing, Communication and Networking Technologies (ICCCNT)*, 2024. (7 citations)
4. E. Vincent, P. Jayant, and A. Chakkan, “DefenSys: An Integrated Platform for Malware Detection and Containerized Attack Simulation Using Deep Learning,” in *Proc. 6th Int. Conf. Smart Electronics and Communication (ICOSEC)*, 2025.
5. A. Ravi, B. Jnyanadeep, M. V. Gagana, P. Jayant, and M. Moharir, “Adaptive ML Framework for SaaS Traffic Classification in Cloud Ecosystem,” in *Proc. Int. Conf. Web Intelligence and Human-Machine Interaction (ICWIHMI)*, 2025.

RESEARCH & PROFESSIONAL EXPERIENCE

Baker Hughes Bangalore, India
Development Engineer Jul 2026 – Present
Digital Technology Intern Jan 2026 – Jul 2026

- Designed and trained a hybrid BERT-CNN natural language processing model for automated document classification, achieving 85% accuracy; investigated architecture choices for combining transformer-based contextual embeddings with convolutional feature extraction, and incorporated human-in-the-loop validation to study model-human agreement, reducing manual audit effort by 50+ hours weekly.
- Engineered a full-stack classification platform (Python, Flask, React, PostgreSQL) with automated message queues, scaling partner intake throughput by 3x across regional enterprise teams.
- Architected production microservices on Microsoft Azure App Service using Microsoft Entra ID RBAC and GitHub Actions CI/CD, reducing deployment cycle times by 40% under a zero-trust security model.

- Designed normalized database schemas and indexed SQL workflows in PostgreSQL, reducing query latency by 35% under 100,000+ concurrent requests for audit-log and export workloads.

Juniper Networks
Software Engineering Intern, Data & Analytics

Bangalore, India
Jul 2024 – Feb 2025

- Engineered a high-throughput Python data processing pipeline handling 1M+ daily network packets, enabling real-time monitoring and automated generation of labeled datasets for downstream security analytics research.
- Built and statistically tuned a microservice classification platform achieving 98% accuracy on network service identification, applying hyperparameter search methodology and optimizing SQL/NoSQL query performance to lower system latency by 25%.
- Established automated unit testing and validation frameworks within an Agile research-and-development workflow, reducing dataset error rates by 30% and maintaining production SLA compliance.

SELECTED RESEARCH PROJECTS

DefenSys: Intelligent Cyber Defense & Threat Platform [GitHub](#)
Tools: C/C++, Python, PyTorch, Docker, Kubernetes, Redis, Linux/Bash

Designed and evaluated an event-driven threat detection platform using statistical anomaly-detection models, with an automated mitigation queue processing 10,000+ simulated requests/sec. Containerized network-traffic simulations with Docker and Kubernetes to benchmark operational capacity and system-health metrics under load.

CoLab: Real-Time Collaborative Distributed Platform [GitHub](#)
Tools: TypeScript, React, Node.js, PostgreSQL, Redis, WebSockets

Architected a low-latency distributed synchronization engine using CRDTs and WebSockets for multi-user state consistency within 50ms response windows. Evaluated PostgreSQL persistence strategies under high-concurrency access patterns.

TECHNICAL SKILLS

Core Programming	C/C++, Python, Java, TypeScript, JavaScript, SQL (Query Optimization, Data Modeling)
Systems & Platform	Linux/Bash, Operating Systems, Computer Networks, Systems Programming, Multi-threading
Cloud & DevOps	Docker, Kubernetes, Linux Kernel/CLI, Microsoft Azure, AWS, GitHub Actions (CI/CD), Git
Backend & Distributed	RESTful APIs, Microservices, Redis Queues, WebSockets, Concurrent Systems Design
AI, ML & Frameworks	PyTorch, TensorFlow, Scikit-learn, Transformers (BERT), Data Pipelines, Telemetry & Observability

HONORS & AWARDS

- **CODE RED’25 Hackathon:** 4th place, Top 1% among 1,000+ teams.
- **ELCIA Next-Gen Hackathon (IIIT Bangalore):** Top 10 for engineering scalable cloud-backed solutions.
- **GDSC-RVCE:** Event Management Lead, Tech Tank Hackathon, overseeing operations and technical infrastructure for 500+ participants.